Single-cell RNA profiling reveals tumor heterogeneity and microenvironment in immune-hot and immune-cold of hepatocellular carcinoma

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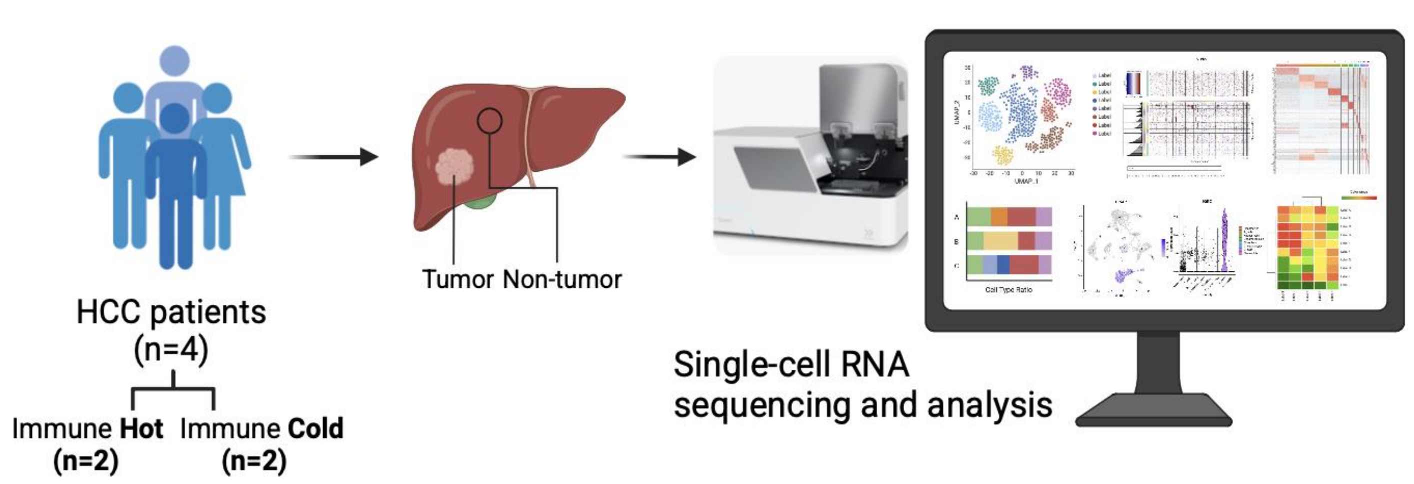
Background

Hepatocellular carcinoma (HCC) is one of the leading causes of cancer-related deaths worldwide and is growing rapidly. The incidence of HCC in the United States has tripled in the past 40 years, with more than 40,000 new cases per year. Although early-stage HCC can be completely cured after surgery, its high recurrence rate and lack of effective treatment led to a 5-year survival rate of only 18%. Tumors can be classified as hot (immunogenic) or cold (non-immunogenic) based on immune cell infiltration, with hot tumors generally more responsive to immunotherapy and a better prognosis. However, the mechanisms that drive and shape these differences in the immune microenvironment between hot and cold tumors remain unclear. Intratumoral factors that affect treatment response are also poorly understood.

Methods

In our previous study, we defined several cohorts of patients with hot tumors and cold tumors. Here, we performed single-cell RNA sequencing (scRNA-seq) in tumor samples from 2 hot cases and 2 cold cases, along with their paired adjacent non-tumor tissues. In total, we analyzed 51,447 high-quality cells from 8 scRNA libraries. scRNA-seq data were then utilized to analyze samples of different immune states, focusing on gene expression and functional changes across various cell types, and their interactions with the tumor. We explore the cell types that have the most significant impact on patients' immune status and their underlying potential mechanisms.

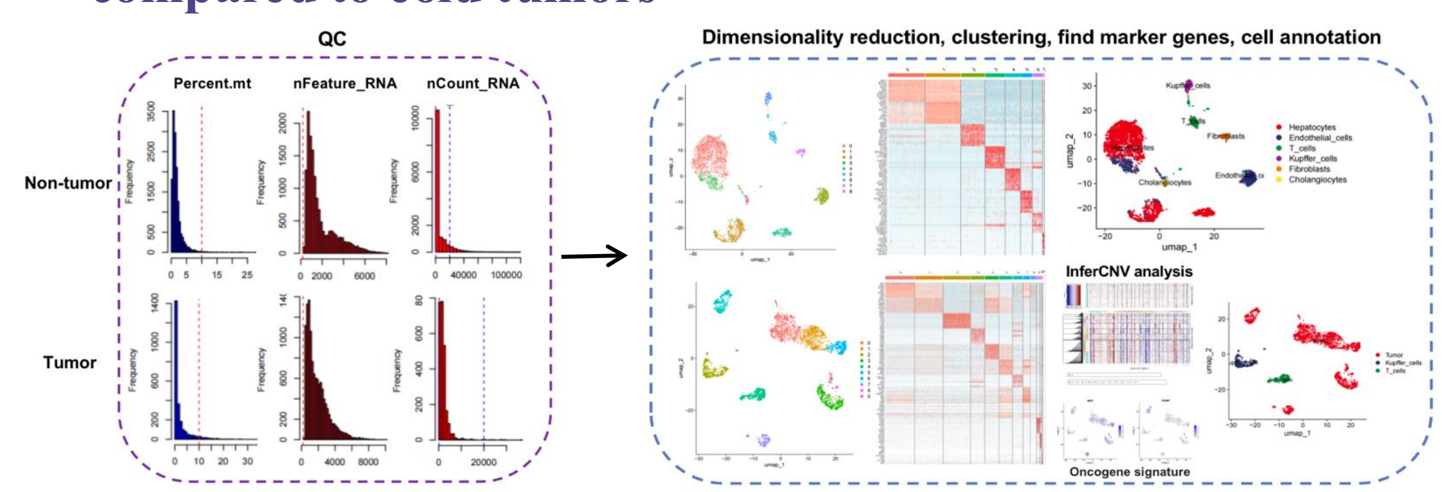
Overview of the study design

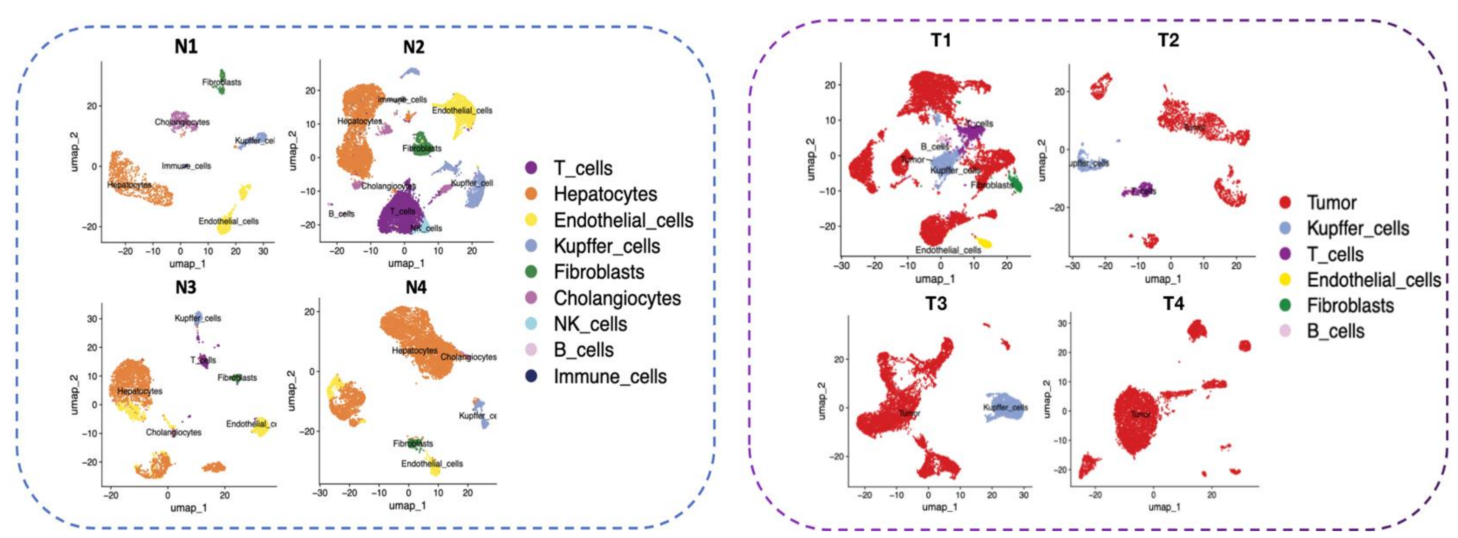


Results

Tumor cells and non-tumor cells were identified based on copy number variation and TCGA gene signature. We found that no T cells were found in cold tumors, while hot tumors had more immune cell infiltration as well as higher activity in immune cell subsets. At the same time, the tumor cells themselves in the hot tumors also showed significant enrichment of immune-related pathways. We compared the differences in T cells between patients with hot and cold samples, and the results showed no clear difference. However, the resident macrophages (Kupffer cells) and neutrophils in patients with hot tumors show remarkable activation of immune pathways such as the IL2-STAT5 pathway, suggesting that they may play a key role in driving the immune thermophenotype.

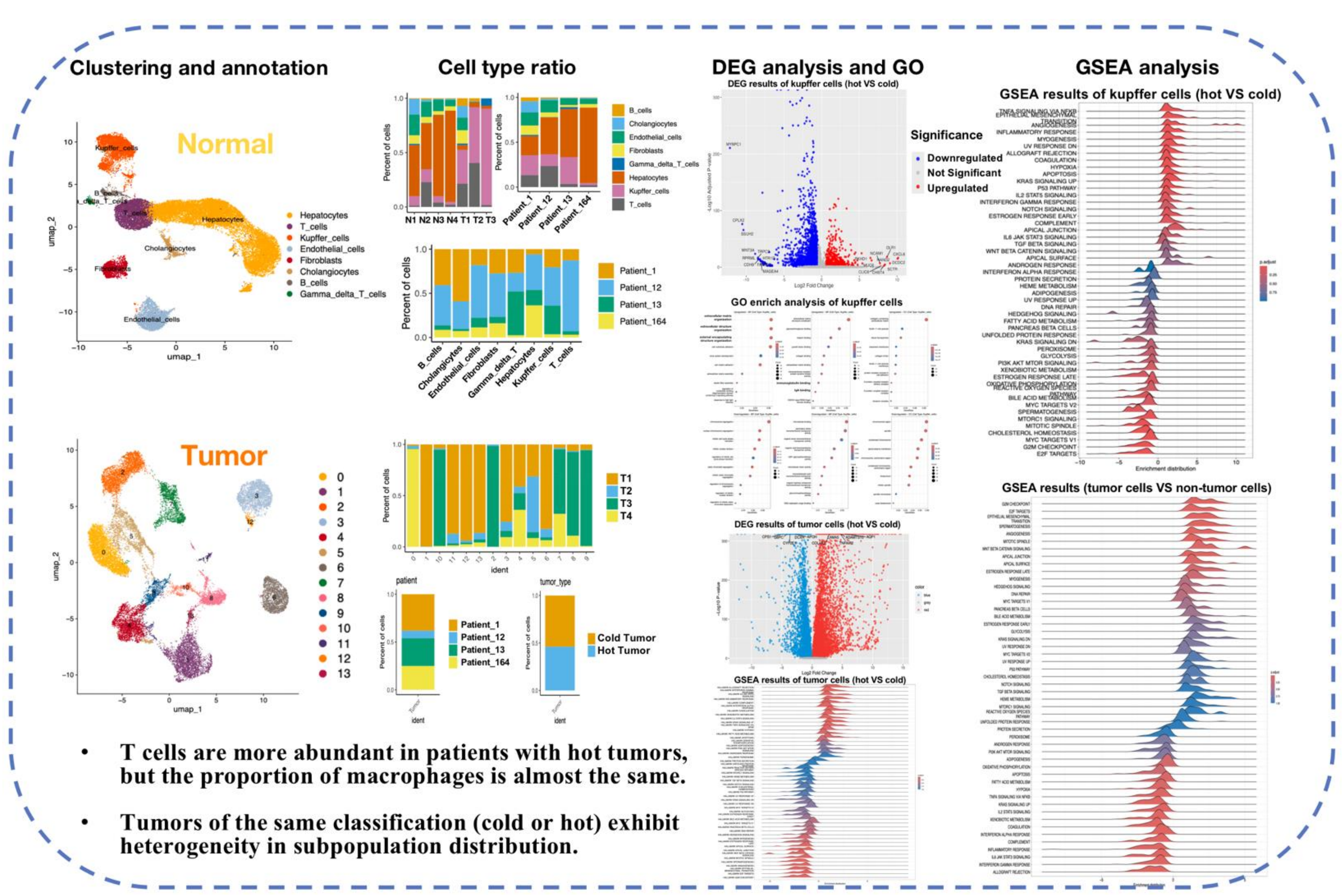
Hot tumors exhibit significantly higher T cell infiltration compared to cold tumors



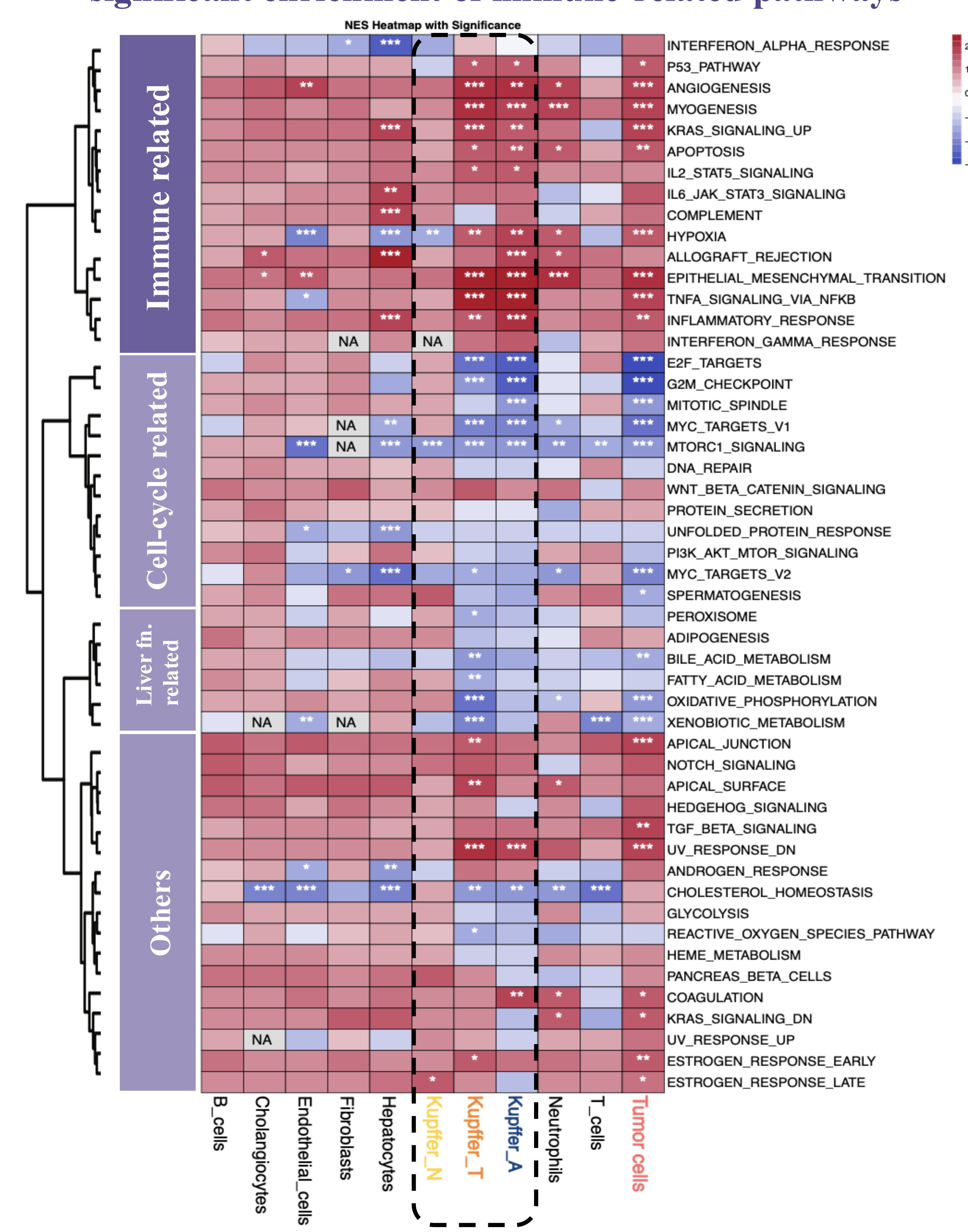


- In order to accurately distinguish between tumor cells and non-tumor cells, we analyzed 8 samples separately. For non-tumor tissues, we manually annotate based on most significant marker genes for each cluster. Tumor cells were identified by inferCNV analysis and oncogene signatures for liver tumor, and the remaining normal cells were manually annotated.

DEG and pathway enrichment analysis reveal elevated immune-related gene expression in macrophages and tumor cells between hot and cold samples



Tumor cells and kupffer cells in hot tumors show significant enrichment of immune-related pathways



- Comparison of individual cell types in patients with hot and cold tumors.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Conclusion

Innate immune cells play a crucial role in shaping the TME. The activation of immune pathways in kupffer cells suggests their potential as therapeutic targets. These insights aid in developing better immunotherapies.

Acknowlegement

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